

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.25)

Metric

$$ds^2 = -\frac{2(r^3x^3 - 3r^2x^2 + 3rx - 1)}{r} du \otimes du - du \otimes dr - r^2 du \otimes dx - dr \otimes du - r^2 dx \otimes du - 2r^2x^3 dx \otimes dx - \frac{r^2}{2x^3} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Metric Functions

$$P_1 = \frac{\sqrt{2}}{2\sqrt{-x^3}}$$

$$P_2 = \frac{\sqrt{2}}{2\sqrt{-x^3 + \frac{3x^2}{r} - \frac{3x}{r^2} + \frac{1}{r^3}}}$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(r^2x^3 - 3rx^2 + 3x - \frac{1}{r}\right) du + (1) dr + (r^2) dx$$

$$m = \left(\sqrt{-x^3}r\right) dx + \left(\frac{ir}{2\sqrt{-x^3}}\right) dy$$

$$\overline{m} = \left(\sqrt{-x^3} r \right) dx + \left(-\frac{i r}{2 \sqrt{-x^3}} \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) du + \left(r^2 x^3 - 3 r x^2 + 3 x - \frac{1}{r} \right) dr$$

$$m = \left(\frac{r \sqrt{-x}}{2 x |x|} \right) dr + \left(-\frac{\sqrt{-x}}{2 r x |x|} \right) dx + \left(-\frac{i x |x|}{r \sqrt{-x}} \right) dy$$

$$\overline{m} = \left(\frac{r \sqrt{-x}}{2 x |x|} \right) dr + \left(-\frac{\sqrt{-x}}{2 r x |x|} \right) dx + \left(\frac{i x |x|}{r \sqrt{-x}} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + r^2 x^3 - 3 r x^2 + 3 x - \frac{1}{r} \partial_r$$

$$\delta = \frac{r \sqrt{-x}}{2 x |x|} \partial_r - \frac{\sqrt{-x}}{2 r x |x|} \partial_x - \frac{i x |x|}{r \sqrt{-x}} \partial_y$$

$$\bar{\delta} = \frac{r \sqrt{-x}}{2 x |x|} \partial_r - \frac{\sqrt{-x}}{2 r x |x|} \partial_x + \frac{i x |x|}{r \sqrt{-x}} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = rx^3 - 3x^2 + \frac{3x}{r} - \frac{1}{r^2}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{-x}}{2x|x|} - \frac{3\sqrt{-x}}{8rx^2|x|}$$

$$\beta = \frac{3\sqrt{-x}}{8rx^2|x|}$$

$$\tau = -\frac{\sqrt{-x}}{2x|x|}$$

$$\pi = \frac{\sqrt{-x}}{2x|x|}$$

$$\epsilon = 0$$

$$\gamma = -rx^3 + \frac{3}{2}x^2 - \frac{1}{2r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{3x^2}{2r} - \frac{3x}{2r^2} + \frac{3}{8rx^4} + \frac{3}{4r^2x^5}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -x^3 + \frac{3x^2}{2r} - \frac{x}{2r^2} + \frac{1}{4x^3} + \frac{3}{8rx^4} + \frac{1}{4r^2x^5}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{x}{r^2} - \frac{1}{r^3} - \frac{1}{2r^2x^5}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"